

Regression Discontinuity Design

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```
bookdown::render_book()
```

Overview

Topics:

1. Continuity Assumption
2. Sharp Regression Discontinuity Design
3. Global and Local Regressions
4. Kernel-Weighted Regressions
5. RDRobust
6. McCrary Density Test

Chapter 1

Continuity Assumption

We will simulate the continuity assumption with fake data. We cannot directly test the continuity assumption, but we will simulate it to see what it looks like.

1.1 Simulate Continuity Assumption

We will set our observation count to 1,000 and set a seed for reproducible results.

```
clear
set obs 1000
set seed 1234567
```

Next, we will generate our fake running variable

```
gen x = rnormal(50, 25)
replace x=0 if x < 0
drop if x > 100
sum x, det
```

(24 real changes made)

(11 observations deleted)

		x
	Percentiles	Smallest
1%	0	0
5%	7.581868	0

10%	17.55611	0	Obs	989
25%	32.95432	0	Sum of wgt.	989
50%	50.69007		Mean	49.34681
		Largest	Std. dev.	23.25304
75%	65.28811	99.0975		
90%	80.07679	99.70067	Variance	540.7037
95%	88.76378	99.97297	Skewness	-.0798667
99%	96.25841	99.97885	Kurtosis	2.44414

Now we will generate the cutoff point at $X = 50$ and when $X \geq 50$ our units of observation are considered treated. We will create a data generation process with $\alpha = 25$ and y is a function of the running variable $1.5 * x$.

```
gen D = 0
replace D = 1 if x > 50
gen y1 = 25 + 0*D + 1.5*x + rnormal(0, 20)
```

Here we have an example of Y^1 not jumping at the cutoff. The potential outcomes are smooth across the discontinuity. Notice that the data generation process had $0 * D$, such that $\delta^{LATE} = 0$

```
twoway (scatter y1 x if D==0, msize(vsmall) msymbol(circle_hollow)) ///
(scatter y1 x if D==1, sort mcolor(blue) msize(vsmall) msymbol(circle_hollow)) ///
(lfit y1 x if D==0, lcolor(red) msize(small) lwidth(medthin) lpattern(solid)) ///
(lfit y1 x, lcolor(dknavy) msize(small) lwidth(medthin) lpattern(solid)), ///
xtitle(Test score (X)) xline(50, legend(off) xlabel(0 "0" 20 "20" 40 "40" ///
50 "Cutoff" 60 "60" 80 "80" 100 "100")) ///
title("Smoothness of Potential Outcome (Y1)")
```

1.2 Simulate the discontinuity

We will set the discontinuity or the $LATE$ equal to 40 or $\delta^{LATE} = 40$. In our prior example, $\delta^{LATE} = 0$ and then replot the data. Now our data generation process will show that there is a discontinuous jump at the cutoff.

```
gen y = 25 + 40*D + 1.5*x + rnormal(0, 20)
scatter y x if D==0, msize(vsmall) || ///
scatter y x if D==1, msize(vsmall) ///
legend(off) xline(50, lstyle(foreground)) || ///
lfit y x if D ==0, color(red) || lfit y x if D ==1, ///
color(red) title("Estimated LATE using simulated data") ///
```

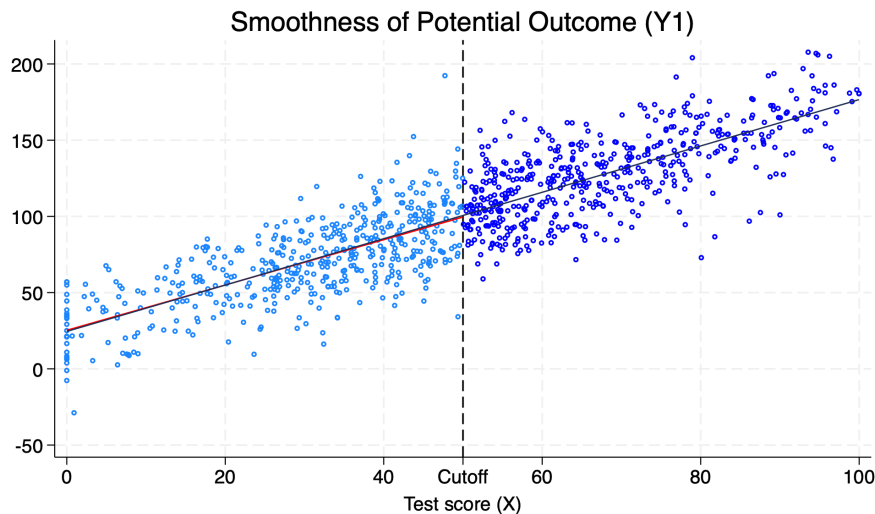



Figure 1.1: Smoothness of Potential Outcomes

```

ytitle("Outcome (Y)")  xtitle("Test Score (X)")  ///
xlabel(0 "0" 20 "20" 40 "40" ///
      50 "Cutoff" 60 "60" 80 "80" 100 "100")

```

We can see that $\delta = 40$ increase in the outcome, Y .

1.3 Nonlinear data generation process

What happens when the data generation process is nonlinear? We need to be careful with the *RDD* design to not impose a linear fit to nonlinear data and assume a discontinuous jump.

We will set the cutoff at $X = 140$, where D is treated if $X > 140$. Due to the functional form, there isn't a discontinuous jump at 140. However, our RDD design it shows that there is a discontinuous jump.

```

set obs 1000
gen x = rnormal(100, 50)
replace x=0 if x < 0
drop if x > 280
sum x, det
gen D = 0
replace D = 1 if x > 140

```

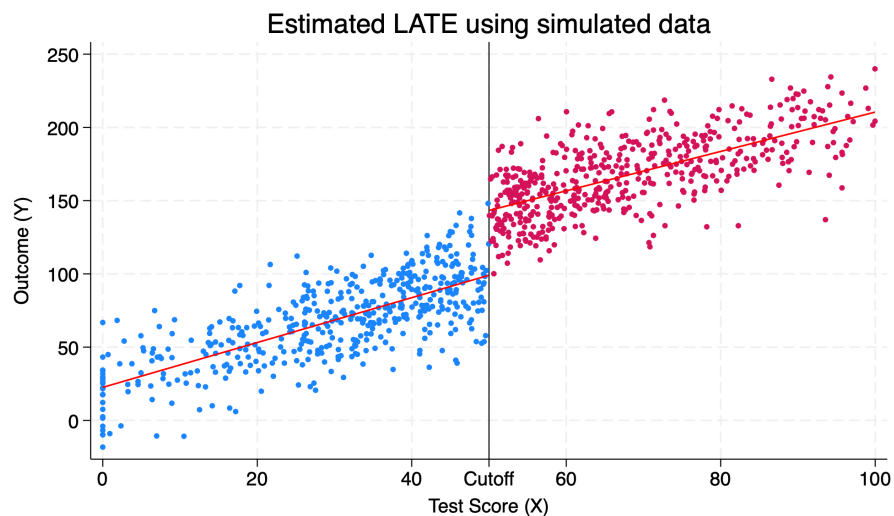


Figure 1.2: A discontinuous jump at the cutoff

```

gen x2 = x*x
gen x3 = x*x*x
gen y = 10000 + 0*D - 100*x +x2 + rnormal(0, 1000)
reg y D x

```

Number of observations (_N) was 0, now 1,000.

(27 real changes made)

(0 observations deleted)

x				

	Percentiles	Smallest		
1%	0	0		
5%	17.31924	0		
10%	31.99439	0	Obs	1,000
25%	65.85717	0	Sum of wgt.	1,000
50%	97.70168		Mean	98.30952
		Largest	Std. dev.	48.48283
75%	131.2771	226.1029		
90%	161.7696	226.2729	Variance	2350.585

95%	179.7875	230.622	Skewness	.0477416
99%	207.1802	239.7431	Kurtosis	2.600393

(201 real changes made)

Source	SS	df	MS	Number of obs	=	1,000
Model	2.6076e+10	2	1.3038e+10	F(2, 997)	=	2093.02
Residual	6.2106e+09	997	6229245.75	Prob > F	=	0.0000
				R-squared	=	0.8076
				Adj R-squared	=	0.8073
Total	3.2286e+10	999	32318766.1	Root MSE	=	2495.8

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]
D	6340.628	278.2154	22.79	0.000	5794.672 6886.583
x	61.58662	2.300816	26.77	0.000	57.07162 66.10162
_cons	4876.251	206.5207	23.61	0.000	4470.986 5281.517

We estimate that the *LATE* is equal to 6340.6. However, we can visualize the incorrect RDD by imposing a linear function form onto a nonlinear function

```
scatter y x if D==0, msize(vsmall) || scatter y x ///
if D==1, msize(vsmall) legend(off) xline(140, ///
lstyle(foreground)) ylabel(none) || lfit y x ///
if D ==0, color(red) || lfit y x if D ==1, ///
color(red) xtitle("Test Score (X)") ///
ytitle("Outcome (Y)") title("Applying Linear Fit to Nonlinear Data") ///
caption("Spurious discontinuity effect due to incorrect specification")
```

We will now model the RDD using a polynomial instead of a linear function.

```
capture drop y
gen y = 10000 + 0*D - 100*x +x2 + rnormal(0, 1000)
reg y D x x2 x3
predict yhat
```

Source	SS	df	MS	Number of obs	=	1,000
Model	3.2207e+10	4	8.0518e+09	F(4, 995)	=	7798.40
Residual	1.0273e+09	995	1032494.42	Prob > F	=	0.0000
				R-squared	=	0.9691
				Adj R-squared	=	0.9690

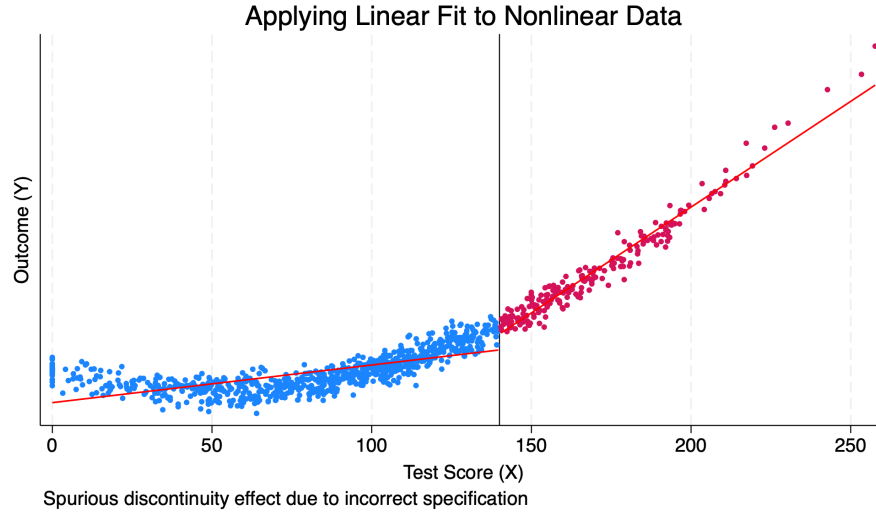


Figure 1.3: Linear Fit to Nonlinear Data

Total | 3.3235e+10 999 33267799.9 Root MSE = 1016.1

	y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
D		44.96491	147.8125	0.30	0.761	-245.0952	335.025
x		-106.7774	5.371053	-19.88	0.000	-117.3173	-96.23752
x2		1.083471	.0586708	18.47	0.000	.9683381	1.198604
x3		-.0002625	.0001779	-1.48	0.141	-.0006117	.0000867
_cons		10081.61	139.5867	72.22	0.000	9807.687	10355.52

(option xb assumed; fitted values)

We fail to reject the null hypothesis that $H_0 : \delta = 0$.

We can visualize the correct functional form, and we can see that the functional form is smooth and continuous.

```
scatter y x if D==0, msize(vsmall) || scatter y x ///
if D==1, msize(vsmall) legend(off) xline(140, ///
lstyle(foreground)) ylabel(none) || line yhat x ///
if D ==0, color(red) sort || line yhat x if D==1, ///
sort color(red) xtitle("Test Score (X)") ///
ytitle("Outcome (Y)") title("Applying Nonlinear Fit to Nonlinear Data") ///
```

```
caption("Polynomial fit shows no discontinuity impact at cutoff")
```

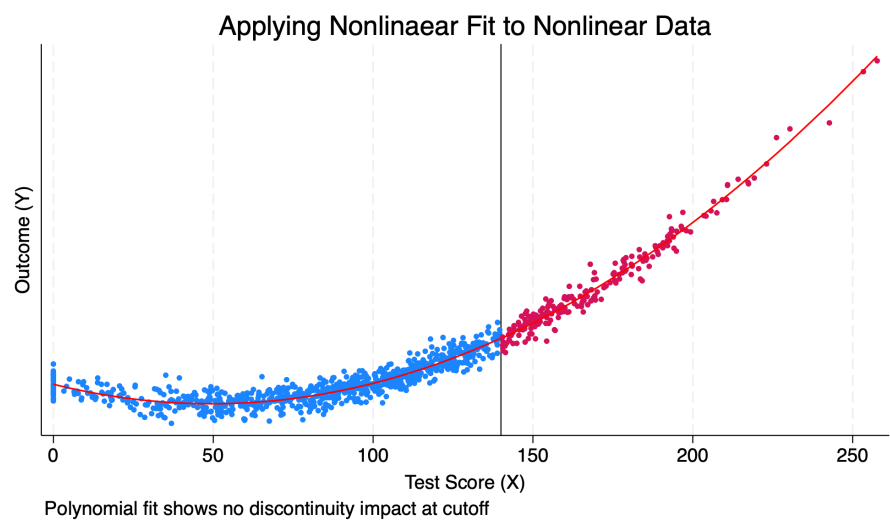


Figure 1.4: Nonlinear fit to nonlinear data

Chapter 2

Sharp RDD

2.1 Replicate Lee, Moretti, and Butler (2004)

Lee, Moretti, and Butler (2004) wanted to test the convergence theory or the divergence theory for policy making. The convergence theory states that candidates will have to moderate their position to appeal to the median voter. The divergence theory states that the winning candidate pursues their preferred policy choices.

2.1.1 Theory

Before an election, or ex ante, voters expect a candidate to choose some policies x^e for democrats and y^e for republicans. Voters expect candidates to win with probability $Pr(x^e, y^e)$.

When $x^e > y^e$, then $\frac{\partial P}{\partial x^e} > 0$ and $\frac{\partial P}{\partial y^e} < 0$.

P^* represents a latent (underlying but not seen) popularity of the democratic party. In other words, P^* is the probability that D would win if policy x is chosen at the democratic policy bliss point c

Convergence Theory: If the convergence theory holds, then $\frac{\partial x^*}{\partial P^L} > 0$. An exogenous shock in x^L would change a change in P^L and the candidate would change their position to satisfy the median voter.

Divergence Theory: If the divergence theory holds, then $\frac{\partial x^L}{\partial P^*} = 0$. Popularity has no affect on policy and voters elect politicians who pursue their preferences and not the voters.

2.1.2 Potential roll-calling voting records outcomes

We are interested in potential roll-call voting records of a candidate following an election. It can be expressed in the following:

$$RC_t = \alpha_0 + \pi_0 P_t^* + \pi_1 D_t + \varepsilon_t$$

Where RC_t is roll-call voting record in time period t , P_t^* is latent popularity of the democratic party in time period t , and D_t is a binary if the democratic candidate won in time period t . In other words, only the winning candidate's voting record is observed.

We are interested in potential roll-call voting records of a candidate following an election in period $t + 1$. It can be expressed in the following:

$$RC_{t+1} = \alpha_0 + \pi_0 P_{t+1}^* + \pi_1 D_{t+1} + \varepsilon_{t+1}$$

Where RC_{t+1} is roll-call voting record in time period $t + 1$, P_{t+1}^* is latent popularity of the democratic party in time period $t + 1$, and D_{t+1} is a binary if the democratic candidate won in time period $t + 1$.

The Problem: We never observe P_t^* and P_{t+1}^* .

If D_t were randomized, then we could estimate an unbiased impact of D_t , but we know that is not the scenario. If this holds then we have the following:

$$\underbrace{E[RC_{t+1}|D_t = 1] - E[RC_{t+1}|D_t = 0]}_{Observable} = \underbrace{\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]}_{Latent} + \underbrace{\pi_1[P_{t+1}^D - P_{t+1}^R]}_{Observable} = \underbrace{\gamma}_{Total\ Effect}$$

Where

1. γ is the total effect of a democratic victory in time t on future policy choice RC_{t+1}
2. $\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]$ is “Affect” component
3. $\pi_1[P_{t+1}^D - P_{t+1}^R]$ is the “Elect” component

Key: if voters merely “elect” policies (under divergence), we should observe little change in the candidate's intended policies following an exogenous increase in the probability in victory. Such that, the “affect” component, $\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]$, will be very small.

What to do: Given that $\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]$ is latent and unobservable, we will back out this part through the parts that are observable: γ, π_1 , and $[P_{t+1}^D - P_{t+1}^R]$

Where the elect component is composed of $\pi_1 * ([P_{t+1}^D - P_{t+1}^R])$ is estimated as the difference between voting records in mean voting records between parties at time t

First, we will estimate the total effect: γ

$$\underbrace{\gamma}_{TotalEffect} = \underbrace{\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]}_{"Affect"} + \underbrace{\pi_1[P_{t+1}^D - P_{t+1}^R]}_{"Elect"}$$

Second, we will estimate the first part of the elect component: π_1

$$\underbrace{E[RC_t|D_t = 1] - E[RC_t|D_t = 0]}_{Observable} = \pi_1$$

Next, we see the second part of the elect component: $P_{t+1}^D - P_{t+1}^R$

$$\underbrace{E[D_{t+1}|D_t = 1] - E[D_{t+1}|D_t = 0]}_{Observable} = P_{t+1}^D - P_{t+1}^R$$

Next, we will use the Americans for Democratic Action (ADA) voting score that is linked to representatives in the US House of Representatives for RC . The roll call is indexed from 0 to 100, where scores closer to 100 are considered more “libreral”.

We will use the vote share as the running variable.

We argue that comes from the exogenous shock in discontinuity in the running variable with a window around the cutoff

2.1.3 Replicate

First, we will replicate Table 1 of Lee, Moretti and Butler (2004) using vote share as our running variable and 50 percent as our cutoff.

We will narrow our window to (0.48, 0.52) for our local regression, and cluster our errors by our *id* variable. Note that our *id* variable is state-district-year identifiers.

First we will estimate our total effect of initial win on future roll calls or γ using a local linear regression with a window of [0.48, 0.52]

```
use "/Users/Sam/Desktop/Econ 672/Data/lmb-data.dta", clear
reg score lagdemocrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id)
```

```

Linear regression                                Number of obs    =      915
                                                F(1, 914)        =     118.98
                                                Prob > F         =      0.0000
                                                R-squared        =     0.1152
                                                Root MSE        =     29.522

```

(Std. err. adjusted for 915 clusters in id)

		Robust				
score	Coefficient	std. err.	t	P> t	[95% conf. interval]	
lagdemocrat	21.28387	1.951234	10.91	0.000	17.45445	25.11329
_cons	31.19576	1.333983	23.39	0.000	28.57773	33.81378

We see a 21.28 increase in the ADA score at the discontinuity, which includes the “elect” and “affect” components.

Next, we will estimate π_1 from the elect component using a local regression using a window of $[0.48, 0.52]$.

```
reg score democrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id)
```

```

Linear regression                                Number of obs    =      915
                                                F(1, 914)        =    1237.69
                                                Prob > F         =      0.0000
                                                R-squared        =     0.5783
                                                Root MSE        =     20.38

```

(Std. err. adjusted for 915 clusters in id)

		Robust				
score	Coefficient	std. err.	t	P> t	[95% conf. interval]	
democrat	47.7056	1.356011	35.18	0.000	45.04434	50.36686
_cons	18.7469	.8432428	22.23	0.000	17.09198	20.40182

Our $\hat{\pi}_1$ is 47.6, which is the increase in ADA scores at time t due to a D treatment/win.

Next, we will calculate $P_{t+1}^D - P_{t+1}^R$

```
reg democrat lagdemocrat if lagdemvoteshare>.48 & lagdemvoteshare<.52, cluster(id)
```

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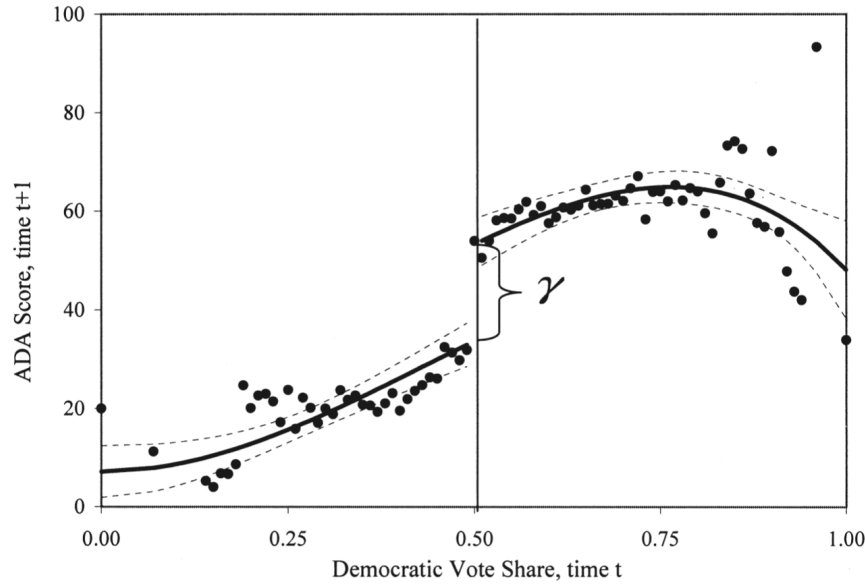


FIGURE I

Total Effect of Initial Win on Future ADA Scores: γ

This figure plots ADA scores after the election at time $t + 1$ against the Democrat vote share, time t . Each circle is the average ADA score within 0.01 intervals of the Democrat vote share. Solid lines are fitted values from fourth-order polynomial regressions on either side of the discontinuity. Dotted lines are pointwise 95 percent confidence intervals. The discontinuity gap estimates

$$\gamma = \underbrace{\pi_0(P_{t+1}^{*D} - P_{t+1}^{*R})}_{\text{"Affect"}} + \underbrace{\pi_1(P_{t+1}^{*D} - P_{t+1}^{*R})}_{\text{"Elect"}}.$$

Figure 2.1: $\hat{\gamma}$ Total effect of initial win on future ADA scores

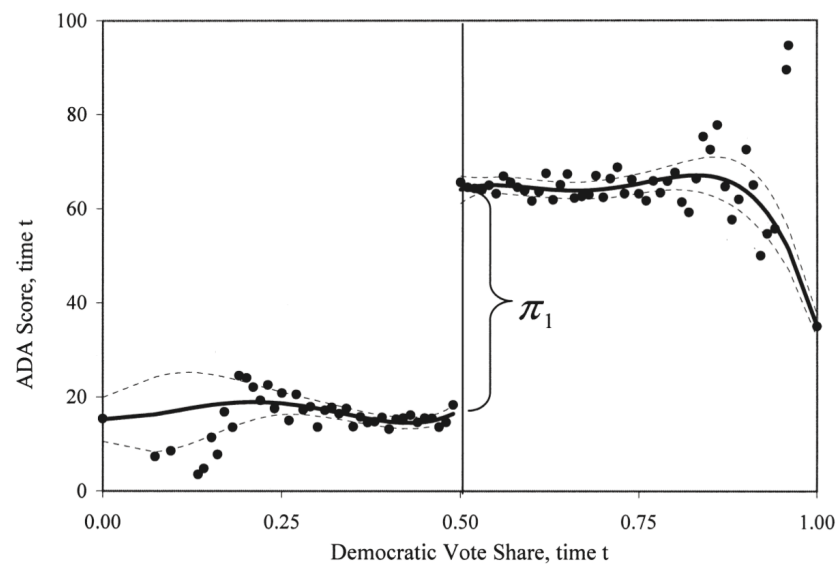


FIGURE IIa
Effect of Party Affiliation: π_1

Figure 2.2: Estimating π_1

Linear regression	Number of obs	=	915
	F(1, 914)	=	280.23
	Prob > F	=	0.0000
	R-squared	=	0.2348
	Root MSE	=	.43767

(Std. err. adjusted for 915 clusters in id)

		Robust				
democrat	Coefficient	std. err.	t	P> t	[95% conf. interval]	
lagdemocrat	.4843287	.0289322	16.74	0.000	.4275475	.5411099
_cons	.2417582	.0200939	12.03	0.000	.2023227	.2811938

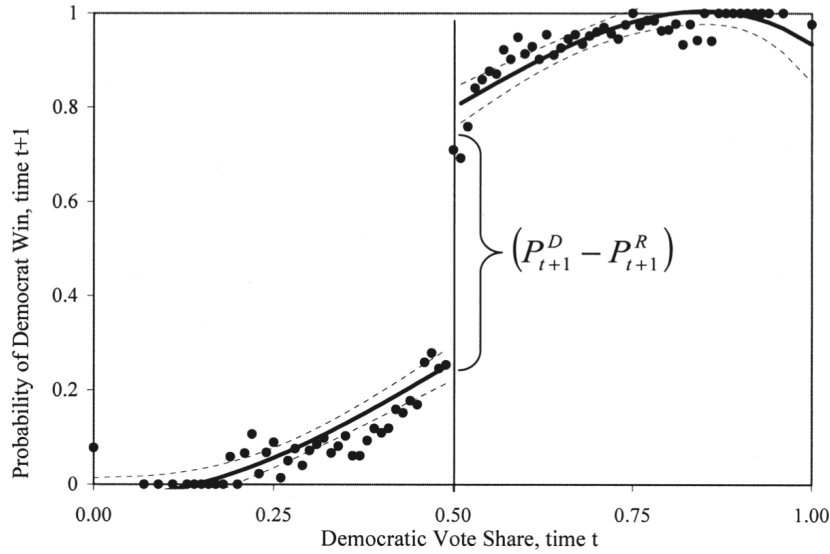


FIGURE IIb

Effect of Initial Win on Winning Next Election: $(P_{t+1}^D - P_{t+1}^R)$

Top panel plots ADA scores after the election at time t against the Democrat vote share, time t . Bottom panel plots probability of Democrat victory at $t + 1$ against Democrat vote share, time t . See caption of Figure III for more details.

Figure 2.3: Estimating $P_{t+1}^D - P_{t+1}^R$

Our $\hat{\pi}_1 * (\widehat{PDt + 1 - PRt + 1})$ is 0.48, which is the effect of initial win on winning the next election.

Now, we have all three of our estimate, so we can back out the “affect” component.

$$\hat{\gamma} - (\hat{\pi}_1 * (PDt + 1 - PRt + 1)) = \hat{\pi}_0$$

The gap at 0.5 comes from π_1 or “elect” and not π_0 or “affect”.

We can see this in the authors’ results.

TABLE I
RESULTS BASED ON ADA SCORES—CLOSE ELECTIONS SAMPLE

Variable	Total effect			Elect component	Affect component
	γ	π_1	$(P_{t+1}^D - P_{t+1}^R)$	$\pi_1[(P_{t+1}^D - P_{t+1}^R)]$	$\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]$
	ADA_{t+1}	ADA_t	DEM_{t+1}	(col. (2))*(col. (3))	(col. (1)) - (col. (4))
	(1)	(2)	(3)	(4)	(5)
Estimated gap	21.2 (1.9)	47.6 (1.3)	0.48 (0.02)	22.84 (2.2)	-1.64 (2.0)

Standard errors are in parentheses. The unit of observation is a district-congressional session. The sample includes only observations where the Democrat vote share at time t is strictly between 48 percent and 52 percent. The estimated gap is the difference in the average of the relevant variable for observations for which the Democrat vote share at time t is strictly between 50 percent and 52 percent and observations for which the Democrat vote share at time t is strictly between 48 percent and 50 percent. Time t and $t + 1$ refer to congressional sessions. ADA_t is the adjusted ADA voting score. Higher ADA scores correspond to more liberal roll-call voting records. Sample size is 915.

Figure 2.4: Table 1 from Lee, Moretti, and Butler (2004)

2.2 Different functional forms of RDD

Now we will replicate Lee, et al. (2004) using different specifications for the RDD. First, we will use all of the data with OLS instead of a local regression without a running variable. Next, we will incorporate a recentered running variable using all of the data. Then, we will use an interaction term between treatment and the running variable using all of the data. For our last global regression, we will use a quadratic interaction term between our running variable and the treatment. Finally, we will use a local regression.

2.2.1 Treatment with no running variable

We will use all of the data instead of just the window around the cutoff. This is the full data set and not a local linear regression.

```
reg score lagdemocrat, cluster(id)
reg score democrat, cluster(id)
reg democrat lagdemocrat, cluster(id)
```

Linear regression	Number of obs	=	13,588
	F(1, 13587)	=	4243.29
	Prob > F	=	0.0000
	R-squared	=	0.2267
	Root MSE	=	28.694

(Std. err. adjusted for 13,588 clusters in id)

	score	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
lagdemocrat		31.50628	.483666	65.14	0.000	30.55823	32.45433
_cons		23.53946	.3374569	69.76	0.000	22.87799	24.20092

Linear regression	Number of obs	=	13,588
	F(1, 13587)	=	9502.26
	Prob > F	=	0.0000
	R-squared	=	0.3756
	Root MSE	=	25.785

(Std. err. adjusted for 13,588 clusters in id)

	score	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
democrat		40.76266	.4181664	97.48	0.000	39.94299	41.58232
_cons		17.5756	.2625459	66.94	0.000	17.06097	18.09022

Linear regression	Number of obs	=	13,588
	F(1, 13587)	=	25739.79
	Prob > F	=	0.0000
	R-squared	=	0.6759
	Root MSE	=	.27929

(Std. err. adjusted for 13,588 clusters in id)

		Robust
--	--	--------

democrat	Coefficient	std. err.	t	P> t	[95% conf. interval]	
lagdemocrat	.8178836	.0050979	160.44	0.000	.807891	.8278761
_cons	.1201058	.0043176	27.82	0.000	.1116428	.1285688

We can see that our discontinuities are quite different under a global regression. However, we do not control the running variable, so we should not put too much stock in these results.

Note: Stata code attributed to Marcelo Perrignon.

2.2.2 Recentered running variable and treatment

Next, we will recenter our running variable and use a global regression. We recenter our 0 to 1 vote share by subtracting by 0.5. So our range becomes $[-0.5, 0.5]$ with 0 being our center.

Recenter running variable of voteshare and rerun regressions

```
gen demvoteshare_c = demvoteshare - 0.5
reg score lagdemocrat demvoteshare_c, cluster(id)
reg score democrat demvoteshare_c, cluster(id)
reg democrat lagdemocrat demvoteshare_c, cluster(id)
```

(11 missing values generated)

Linear regression	Number of obs	=	13,577
	F(2, 13576)	=	2115.71
	Prob > F	=	0.0000
	R-squared	=	0.2274
	Root MSE	=	28.686

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
score	Coefficient	std. err.	t	P> t	[95% conf. interval]	
lagdemocrat	33.45118	.8481987	39.44	0.000	31.7886	35.11377
demvoteshare_c	-5.62558	1.898081	-2.96	0.003	-9.346082	-1.905077
_cons	22.88345	.4433244	51.62	0.000	22.01447	23.75242

Linear regression	Number of obs	=	13,577
	F(2, 13576)	=	6192.13
	Prob > F	=	0.0000
	R-squared	=	0.4242
	Root MSE	=	24.764

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
	score	Coefficient	std. err.	t	P> t	[95% conf. interval]
democrat		58.50236	.65638	89.13	0.000	57.21576 59.78895
demvoteshare_c		-48.93761	1.641489	-29.81	0.000	-52.15516 -45.72006
_cons		11.03413	.3363049	32.81	0.000	10.37493 11.69334

Linear regression	Number of obs	=	13,577
	F(2, 13576)	=	48967.17
	Prob > F	=	0.0000
	R-squared	=	0.7354
	Root MSE	=	.25242

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
	democrat	Coefficient	std. err.	t	P> t	[95% conf. interval]
lagdemocrat		.5516115	.0103234	53.43	0.000	.5313761 .5718469
demvoteshare_c		.7725248	.0187564	41.19	0.000	.7357597 .8092898
_cons		.2116664	.0052743	40.13	0.000	.2013279 .2220048

Our results are closer to Lee, et al. (2004), but they are still far off. Please note that we keep the function form of the slope for the running variable the same on both sides of the cutoff.

Note: Stata code attributed to Marcelo Perrailon.

2.2.3 Interaction between treatment and running variable

We are going to allow for intercepts and slopes to differ with interacting categorical and continuous variables. Why do we want to do this? We allow for the slope to differ on both sides of the discontinuity. As seen above in Figure 1, the slopes may differ on each side of the discontinuity.

Please note that Cunningham likes to use the *xi:*, while I prefer to use the interaction operators *##*.

```
xi: reg score i.lagdemocrat*demvoteshare_c, cluster(id)
xi: reg score i.democrat*demvoteshare_c, cluster(id)
xi: reg democrat i.lagdemocrat*demvoteshare_c, cluster(id)
```

```
i.lagdemocrat      _Ilagdemocr_0-1      (naturally coded; _Ilagdemocr_0 omitted)
i.lagd~t*demv~c    _IlagXdemvo_#        (coded as above)
```

Linear regression	Number of obs	=	13,577
	F(3, 13576)	=	1863.06
	Prob > F	=	0.0000
	R-squared	=	0.2669
	Root MSE	=	27.944

(Std. err. adjusted for 13,577 clusters in id)

	score	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
_Ilagdemocr_1		30.50803	.8172343	37.33	0.000	28.90614	32.10993
demvoteshare_c		66.04181	3.159572	20.90	0.000	59.84861	72.23501
_IlagXdemvo_1		-96.47474	3.851734	-25.05	0.000	-104.0247	-88.92481
_cons		31.43518	.5409599	58.11	0.000	30.37482	32.49554

```
i.democrat      _Idemocrat_0-1      (naturally coded; _Idemocrat_0 omitted)
i.demo~t*demv~c  _IdemXdemvo_#        (coded as above)
```

Linear regression	Number of obs	=	13,577
	F(3, 13576)	=	4160.73
	Prob > F	=	0.0000
	R-squared	=	0.4344
	Root MSE	=	24.544

(Std. err. adjusted for 13,577 clusters in id)

	score	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
_Idemocrat_1		55.43136	.6373768	86.97	0.000	54.18201	56.68071
demvoteshare_c		-5.682785	2.609124	-2.18	0.029	-10.79703	-.5685407
_IdemXdemvo_1		-55.15188	3.217652	-17.14	0.000	-61.45893	-48.84484
_cons		16.81598	.4184826	40.18	0.000	15.9957	17.63627

```

i.lagdemocrat      _llagdemocr_0-1      (naturally coded; _llagdemocr_0 omitted)
i.lagd~t*demv~c    _llagXdemvo_#      (coded as above)

```

```

Linear regression      Number of obs      =      13,577
                      F(3, 13576)         =      25212.81
                      Prob > F             =      0.0000
                      R-squared            =      0.7489
                      Root MSE           =      .2459

```

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
democrat	Coefficient	std. err.	t	P> t	[95% conf. interval]	
_llagdemocr_1	.5257231	.0104514	50.30	0.000	.505237	.5462092
demvoteshare_c	1.402922	.0443487	31.63	0.000	1.315993	1.489852
_llagXdemvo_1	-.8486068	.0490015	-17.32	0.000	-.9446565	-.7525571
_cons	.2868887	.0077506	37.02	0.000	.2716965	.3020809

Note: *xi* is interaction expansion: <https://www.stata.com/manuals13/rxi.pdf>

Note: Stata code attributed to Marcelo Perrailon.

You can use `##` between two variables for interactions as well (I prefer this method). You use `i.varname` for the categorical and `c.varname` for the continuous. So we are interacting a binary variable with a continuous one here to allow for different slopes and intercepts.

```

reg score i.lagdemocrat##c.demvoteshare_c, cluster(id)
reg score i.democrat##c.demvoteshare_c, cluster(id)
reg democrat i.lagdemocrat##c.demvoteshare_c, cluster(id)

```

```

Linear regression      Number of obs      =      13,577
                      F(3, 13576)         =      1863.06
                      Prob > F             =      0.0000
                      R-squared            =      0.2669
                      Root MSE           =      27.944

```

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
score	Coefficient	std. err.	t	P> t	[95% conf. interval]	

1.lagdemocrat		30.50803	.8172343	37.33	0.000	28.90614	32.10993
demvoteshare_c		66.04181	3.159572	20.90	0.000	59.84861	72.23501
lagdemocrat#							
c.							
demvoteshare_c							
1		-96.47474	3.851734	-25.05	0.000	-104.0247	-88.92481
_cons		31.43518	.5409599	58.11	0.000	30.37482	32.49554

Linear regression

Number of obs = 13,577
 F(3, 13576) = 4160.73
 Prob > F = 0.0000
 R-squared = 0.4344
 Root MSE = 24.544

(Std. err. adjusted for 13,577 clusters in id)

score		Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
1.democrat		55.43136	.6373768	86.97	0.000	54.18201 56.68071
demvoteshare_c		-5.682785	2.609124	-2.18	0.029	-10.79703 -.5685407
democrat#						
c.						
demvoteshare_c						
1		-55.15188	3.217652	-17.14	0.000	-61.45893 -48.84484
_cons		16.81598	.4184826	40.18	0.000	15.9957 17.63627

Linear regression

Number of obs = 13,577
 F(3, 13576) = 25212.81
 Prob > F = 0.0000
 R-squared = 0.7489
 Root MSE = .2459

(Std. err. adjusted for 13,577 clusters in id)

democrat		Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
----------	--	-------------	---------------------	---	------	----------------------

1.lagdemocrat		.5257231	.0104514	50.30	0.000	.505237	.5462092
demvoteshare_c		1.402922	.0443487	31.63	0.000	1.315993	1.489852
lagdemocrat#							
c.							
demvoteshare_c							
1		-.8486068	.0490015	-17.32	0.000	-.9446565	-.7525571
_cons		.2868887	.0077506	37.02	0.000	.2716965	.3020809

With our full-sample interaction, we are starting to get closer to our results from the local linear regression and the results Lee, Moretti, and Butler (2004).

2.2.4 Quadratic interaction between treatment and running variable

Next, we will use a global linear regression using an quadratic interaction. Why do we want to use a quadratic interaction? Not only do we allow the slopes to differ, we also allow for flexibility in the function form. This will better match the data on both sides of the discontinuity. Furthermore, if we misspecify the functional form, then we may attribute a discontinuity when one does not exist.

```
gen demvoteshare_sq = demvoteshare_c^2
reg score i.lagdemocrat##c.(demvoteshare_c demvoteshare_sq), cluster(id)
reg score i.democrat##c.(demvoteshare_c demvoteshare_sq), cluster(id)
reg democrat i.lagdemocrat##c.(demvoteshare_c demvoteshare_sq), cluster(id)
```

Linear regression	Number of obs	=	13,577
	F(5, 13576)	=	1529.38
	Prob > F	=	0.0000
	R-squared	=	0.3707
	Root MSE	=	25.892

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
score	Coefficient	std. err.	t	P> t	[95% conf. interval]	
1.lagdemocrat	13.02968	1.268439	10.27	0.000	10.54337	15.516
demvoteshare_c	134.9765	9.634337	14.01	0.000	116.0919	153.8611
demvoteshare_sq	212.1269	22.39469	9.47	0.000	168.2302	256.0236
lagdemocrat#						

c.							
demvoteshare_c							
1		57.05469	15.15178	3.77	0.000	27.35509	86.7543
lagdemocrat#							
c.							
demvoteshare_sq							
1		-641.8509	30.80241	-20.84	0.000	-702.2279	-581.4739
_cons		33.54745	.7068697	47.46	0.000	32.16189	34.93302

Linear regression

Number of obs = 13,577
 F(5, 13576) = 2589.02
 Prob > F = 0.0000
 R-squared = 0.4559
 Root MSE = 24.075

(Std. err. adjusted for 13,577 clusters in id)

score		Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.democrat		44.40229	.9086222	48.87	0.000	42.62126	46.18331
demvoteshare_c		-23.8496	6.710943	-3.55	0.000	-37.00398	-10.69522
demvoteshare_sq		-41.72917	14.67192	-2.84	0.004	-70.48817	-12.97018
democrat#							
c.							
demvoteshare_c							
1		111.8963	9.779383	11.44	0.000	92.72733	131.0652
democrat#							
c.							
demvoteshare_sq							
1		-229.9544	19.53577	-11.77	0.000	-268.2472	-191.6615
_cons		15.60635	.5747061	27.16	0.000	14.47984	16.73285

Linear regression

Number of obs = 13,577
 F(5, 13576) = 89733.30
 Prob > F = 0.0000
 R-squared = 0.8220

Root MSE = .20703

(Std. err. adjusted for 13,577 clusters in id)

		Robust				
democrat	Coefficient	std. err.	t	P> t	[95% conf. interval]	
1.lagdemocrat	.3216837	.0181913	17.68	0.000	.2860263	.3573411
demvoteshare_c	2.798344	.1928588	14.51	0.000	2.420314	3.176374
demvoteshare_sq	4.294014	.4472409	9.60	0.000	3.41736	5.170668
lagdemocrat#c.						
1	.0909402	.2371611	0.38	0.701	-.3739284	.5558089
lagdemocrat#c.						
demvoteshare_sq						
1	-8.804368	.5077003	-17.34	0.000	-9.799531	-7.809205
_cons	.3296468	.0125535	26.26	0.000	.3050402	.3542534

With our quadratic interaction, we start to move away from our local linear regression (except for the estimate of π_1). Our estimates of γ and $(PDt + 1 - PRt + 1)$ are far off from where we expect them to be.

2.2.5 Local regression: 5 points from cutoff with OLS

Finally, we will use a local linear regression with a quadratic interaction. With our local linear regression with a quadratic interaction, we narrow our window, but we allow for a more flexible functional form on both sides of the discontinuity.

```
reg score i.lagdemocrat##c.(demvoteshare_c demvoteshare_sq) if lagdemvoteshare>.45 & lagdemvoteshare<.45
reg score i.democrat##c.(demvoteshare_c demvoteshare_sq) if lagdemvoteshare>.45 & lagdemvoteshare<.45
reg democrat i.lagdemocrat##c.(demvoteshare_c demvoteshare_sq) if lagdemvoteshare>.45 & lagdemvoteshare<.45
```

Linear regression

Number of obs = 2,441
 F(5, 2440) = 261.93
 Prob > F = 0.0000
 R-squared = 0.4048
 Root MSE = 23.951

(Std. err. adjusted for 2,441 clusters in id)

score	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.lagdemocrat	3.967238	1.490509	2.66	0.008	1.044445	6.890032
demvoteshare_c	241.8926	20.54643	11.77	0.000	201.6023	282.1828
demvoteshare_sq	407.0713	62.42092	6.52	0.000	284.6678	529.4748
lagdemocrat#						
c.						
demvoteshare_c						
1	27.51541	28.75342	0.96	0.339	-28.86823	83.89905
lagdemocrat#						
c.						
demvoteshare_sq						
1	-1002.938	79.32636	-12.64	0.000	-1158.492	-847.3844
_cons	40.65686	.9475232	42.91	0.000	38.79883	42.51489

Linear regression

Number of obs = 2,441
 F(5, 2440) = 717.83
 Prob > F = 0.0000
 R-squared = 0.5801
 Root MSE = 20.117

(Std. err. adjusted for 2,441 clusters in id)

score	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.democrat	46.87991	1.542781	30.39	0.000	43.85462	49.90521
demvoteshare_c	-27.71422	15.03124	-1.84	0.065	-57.18953	1.761092
demvoteshare_sq	-87.87141	29.14324	-3.02	0.003	-145.0195	-30.72336
democrat#						
c.						
demvoteshare_c						
1	78.48126	23.09883	3.40	0.001	33.18592	123.7766
democrat#						
c.						

demvoteshare_sq							
1		-132.0792	50.97886	-2.59	0.010	-232.0455	-32.11286
_cons		17.51743	1.013373	17.29	0.000	15.53027	19.50459

Linear regression

Number of obs = 2,441
F(5, 2440) = 1642.94
Prob > F = 0.0000
R-squared = 0.7005
Root MSE = .27372

(Std. err. adjusted for 2,441 clusters in id)

		Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.lagdemocrat		.1181676	.0203498	5.81	0.000	.0782629	.1580723
demvoteshare_c		5.095347	.3027069	16.83	0.000	4.501758	5.688936
demvoteshare_sq		9.265549	.9176752	10.10	0.000	7.466046	11.06505
lagdemocrat#							
c.							
demvoteshare_c							
1		.0359484	.38651	0.09	0.926	-.7219732	.79387
lagdemocrat#							
c.							
demvoteshare_sq							
1		-18.30185	1.05566	-17.34	0.000	-20.37193	-16.23176
_cons		.4411991	.0135581	32.54	0.000	.4146126	.4677855

Interestingly, our local linear regression with a quadratic interaction does much worse for estimating γ and $(PDt + 1 - PRt + 1)$

Chapter 3

Visualize Global and Local Regressions

We'll now switch to local polynomial regressions as a way to assess the impact of treatment at the cutoff. A nonparametric method in RDD is to estimate a model that doesn't assume a functional form for the relationship between the outcome and running variable:

$$y = f(x) + \varepsilon$$

We'll calculate the $E[Y]$ for each bin of the running variable x . Stata has *cmogram* command to estimate nonparametric graphics.

3.1 Install *cmogram*

One method to help look for discontinuities and the **continuity assumption** is *cmogram*. It is good to visualize what potential trends in the running variable are - e.g. your eyes.

First, we will need to install *cmogram*.

```
ssc install cmogram //If this gives you an error, please comment it out and install cmogram manually
```

Make sure to check the bins after *cmogram* has run in the output window with *cmogram*. If there doesn't seem to be any trends in the running variable, then polynomials will not help much.

3.2 Global Regression with Quadratic Fit

Next, we can visualize our different functional forms.

```
cmogram score lagdemvoteshare, cut(0.5) scatter line(0.5) qfitci title("Global with Quadratic Fit")
```

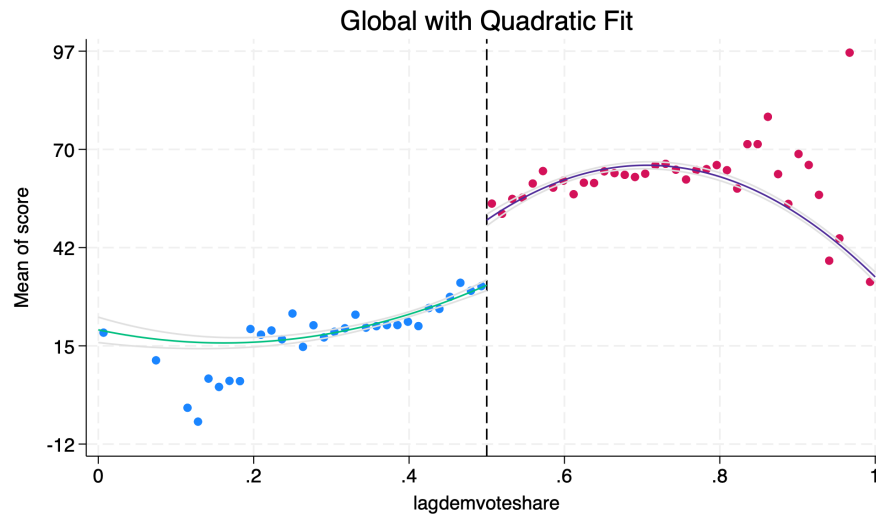


Figure 3.1: Global regression with Quadratic Fit

This is similar to Lee, et al. (2004) Figure 1, but with slightly different bin sizes. A quadratic fit seem to mimic Lee, Moretti, and Butler (2004) the closest.

3.3 Local Regression with Quadratic Fit

Next, we will use a local regression with a window of (0.4,0.6) and use a quadratic fit.

```
cmogram score lagdemvoteshare if lagdemvoteshare > 0.4 & lagdemvoteshare < 0.6, cut(0.5)
```

The quadratic fit might not be appropriate here, which is maybe why our estimates were so far off.

3.4 Global Regression with Linear Fit

Next, we will focus on a global linear regression.

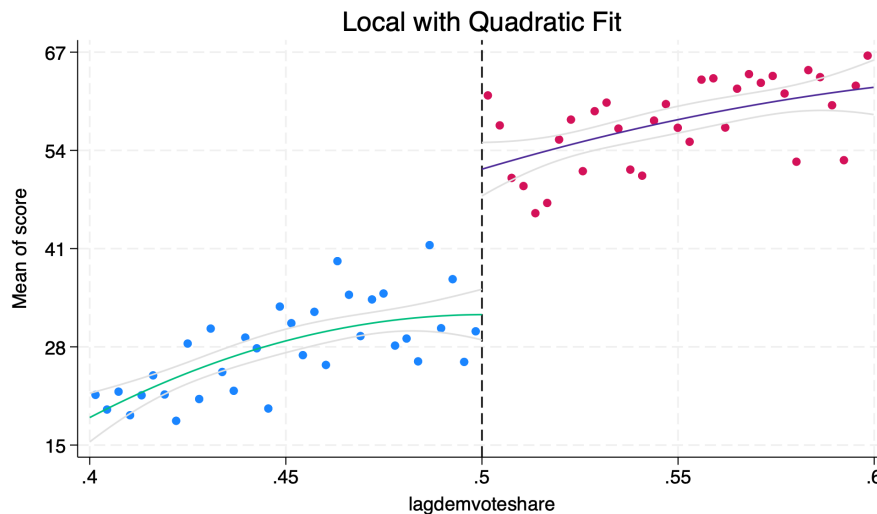


Figure 3.2: Local regression with Quadratic Fit

```
cmogram score lagdemvoteshare, cut(0.5) scatter line(0.5) lfit title("Global with Linear Fit")
```

Linear Fit seem to be influenced by outliers far from the cutoff. The global linear fit may work around $[0.2, 0.8]$, but does not perform well below 0.2 and 0.8.

3.5 Local Regression with Linear Fit

Next, we will use a local linear regression with a window of $(0.4, 0.6)$.

```
cmogram score lagdemvoteshare if lagdemvoteshare > 0.4 & lagdemvoteshare < 0.6, cut(0.5) scatter
```

The local linear fit appears to be appropriate. Using this method, we find our results were closest to Lee, et al. (2004).

3.6 Global Regression with Lowess Fit

Finally, we will use a global regression but we will use a lowess fit. Lowess comes from “locally weighted scatter plot smooth”. A lowess fit is a non-parametric regression method that is particularly useful for creating a smooth line through a scatterplot of data points.

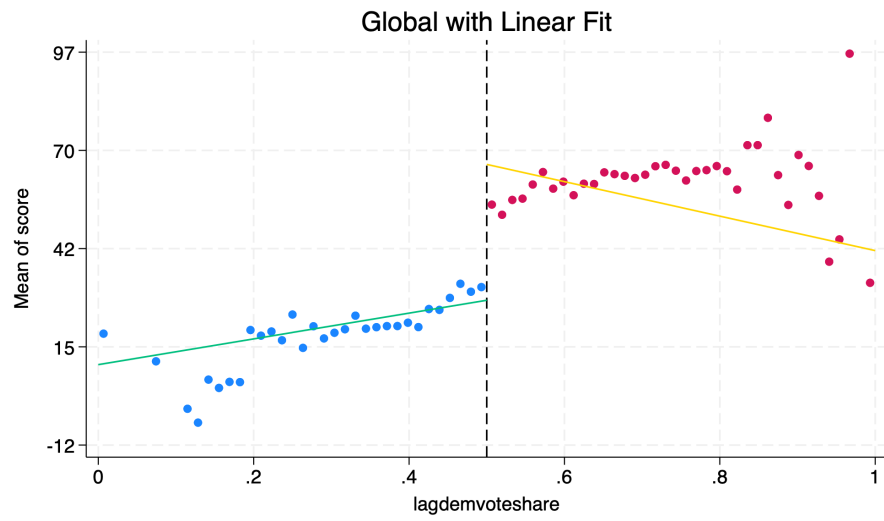


Figure 3.3: Global Linear Regression

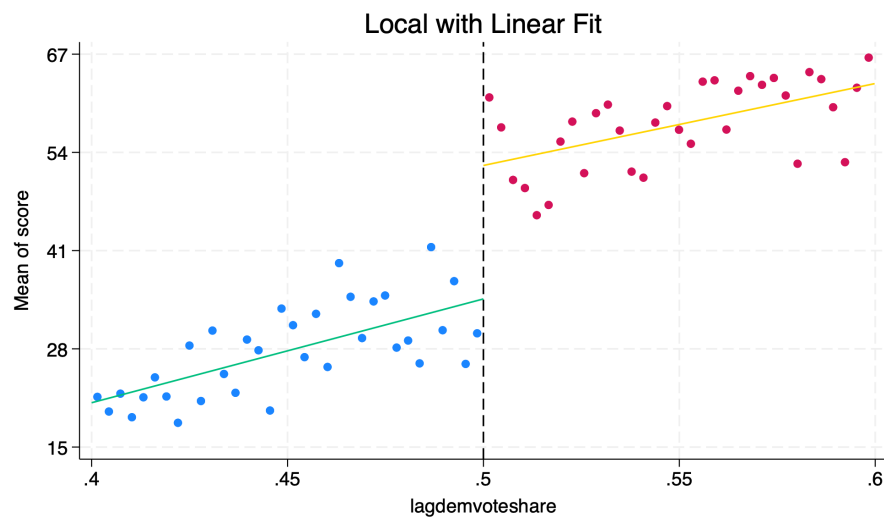


Figure 3.4: Local Linear Regression

```
cmogram score lagdemvoteshare, cut(0.5) scatter line(0.5) lowess title ("Global with Lowess Fit")
```

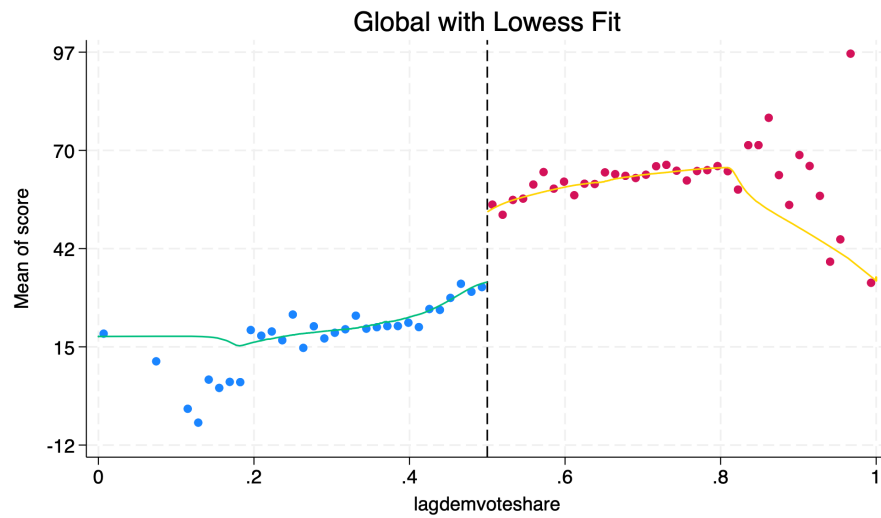


Figure 3.5: Global Regression with a Lowess Fit

Chapter 4

Kernel-weighted local polynomial regression

Kernel-weighted local polynomial regressions are a weighted regression restricted to a window-like (local) where the chosen kernel provides the weights. Observations closer to the cutoff have greater weight, and you need to select the bandwidth window. The method is sensitive to the size of the bandwidth window

We use a triangle kernel with the function *lpoly*. We need to specify the options *kernel(triangle)* and we need to specify the bandwidth size with (*bwidth*)

```
capture drop sdem* x1 x0
lpoly score demvoteshare if democrat == 0, nograph kernel(triangle) gen(x0 sdem0) bwidth(0.1)}
lpoly score demvoteshare if democrat == 1, nograph kernel(triangle) gen(x1 sdem1) bwidth(0.1)}
scatter sdem1 x1, color(red) msize(small) || scatter sdem0 x0, msize(small) color(red) ///
xline(0.5,lstyle(dot)) legend(off) xtitle("Democratic vote share") ytitle("ADA score")
```

We estimate the local polynomial *LATE* using the following:

```
gen diff = sdem1 - sdem0
list sdem1 sdem0 diff in 1/1
```

(13,538 missing values generated)

	sdem1	sdem0	diff
1.	64.39525	16.247676	48.14758

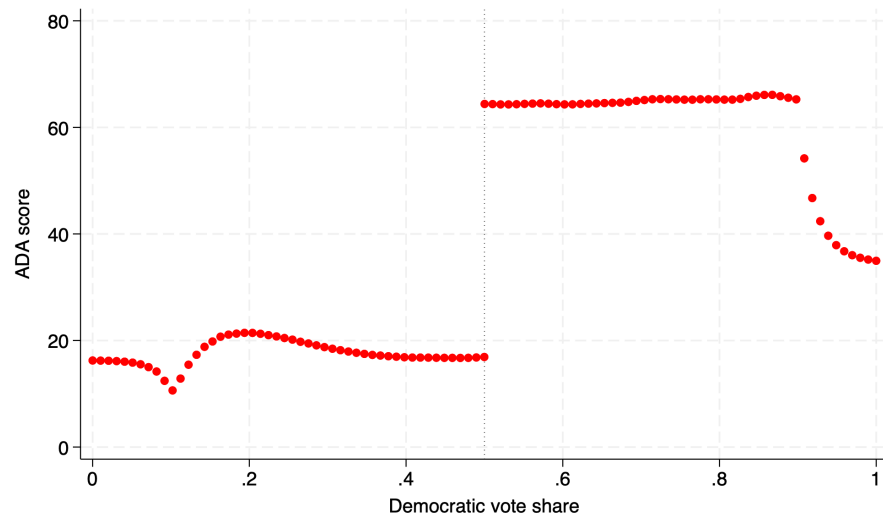


Figure 4.1: Kernel-weighted local polynomial regression

Stata code attributed to Marcelo Perrignon.

Chapter 5

RDRobust

There is a bias-variance tradeoff when selecting bandwidth size. The smaller the bandwidth window, the lower the bias, but fewer observations which increases the variance. The larger the window, the higher the bias, but more observations, which decreases the variance. *RDRobust* optimizes the tradeoff by choosing the optimal bandwidth sizes *Which may vary to the left or right of the cutoff.

Use local polynomial point estimators with bias correction

```
ssc install rdrobust
```

We can estimate the *LATE* with the *RDRobust* command.

```
rdrobust score demvoteshare, c(0.5)
```

Mass points detected in the running variable.

Sharp RD estimates using local polynomial regression.

	Cutoff c = .5 Left of c	Right of c		
-----+-----			Number of obs =	13577
Number of obs	5480	8097	BW type =	mserd
Eff. Number of obs	2112	1895	Kernel =	Triangular
Order est. (p)	1	1	VCE method =	NN
Order bias (q)	2	2		
BW est. (h)	0.086	0.086		
BW bias (b)	0.141	0.141		
rho (h/b)	0.609	0.609		
Unique obs	2770	3351		

Outcome: score. Running variable: demvoteshare.

Method	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
Conventional	46.491	1.2405	37.4775	0.000	44.06	48.9227
Robust	-	-	31.4254	0.000	43.2929	49.0524

Estimates adjusted for mass points in the running variable.

Chapter 6

McCrary Density Test

We can use local polynomial density estimations with *RDDensity* and *LPDensity* commands. The McCrary density test can test to see if there were any manipulations in the running variable at the cutoff. Our null hypothesis for the McCrary density test is no manipulation around the cutoff.

We want to make sure that the following are not occurring:

1. The assignment rule is known in advance
2. Economic agents are interested in adjusting
3. Economic agents have time to adjust
4. The cutoff is not exogenous (or is endogenous with other factors)
5. There is non-random heaping along the running variable

When we look at arbitrary rules known in advanced, such as the FMLA cutoff at 50 employees, we see a pileup or heaping of firms with 49 employees. Such that, the density of firms with 49 employees is much greater than the density of firms with 50 employees. Firms are economic agents self-selecting into 49 employees to avoid regulation.

```
net install rddensity, from(https://raw.githubusercontent.com/rdpackages/rddensity/master/stata)
net install lpdensity, from(https://raw.githubusercontent.com/nppackages/lpdensity/master/stata)
```

```
rddensity demvoteshare, c(0.5) plot
```

Computing data-driven bandwidth selectors.

Point estimates and standard errors have been adjusted for repeated observations.
(Use option `nomasspoints` to suppress this adjustment.)

RD Manipulation test using local polynomial density estimation.

c =	0.500	Left of c	Right of c	Number of obs =	13577
-----+-----				Model =	unrestricted
Number of obs	5480		8097	BW method =	comb
Eff. Number of obs	1994		2250	Kernel =	triangular
Order est. (p)	2		2	VCE method =	jackknife
Order bias (q)	3		3		
BW est. (h)	0.081		0.103		

Running variable: demvoteshare.

-----+-----		
Method	T	P> T
-----+-----		
Robust	0.3628	0.7168
-----+-----		

P-values of binomial tests. (H0: prob = .5)

-----+-----			
Window Length / 2	<c	>=c	P> T
-----+-----			
0.000	10	11	1.0000
0.001	16	23	0.3368
0.001	31	34	0.8043
0.002	43	36	0.4999
0.002	55	43	0.2664
0.003	63	58	0.7163
0.003	74	71	0.8682
0.004	84	77	0.6364
0.004	90	86	0.8212
0.004	96	97	1.0000
-----+-----			

We fail to reject the null hypothesis, and we do not see evidence of manipulation around the cutoff.

Stata code attributed to Marcelo Perrignon.

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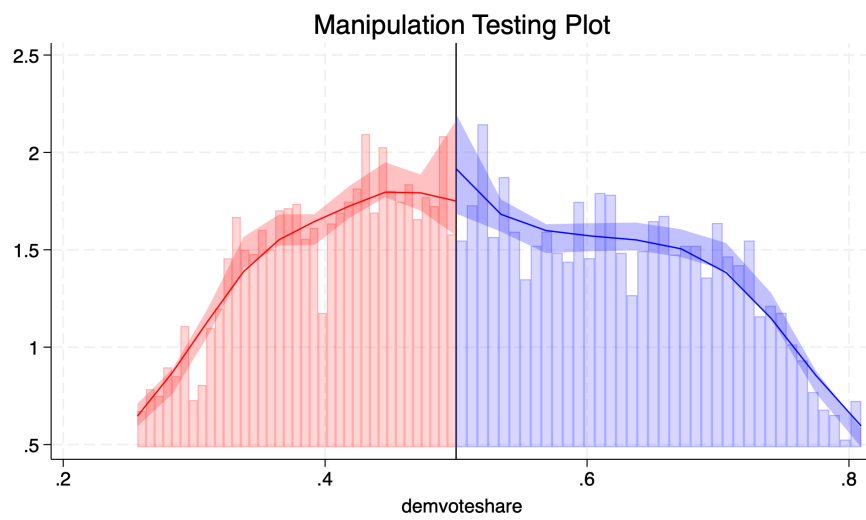


Figure 6.1: McCrary Density Test for Manipulation